



C Programming Pu - practice questions

programming in c (Pokhara University)



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Pokhara University C Programming Question Bank

Structures and File Handling Programs

1. Input and Display Student Records Using Structure

```
#include <stdio.h>
struct Student {
    int roll;
    char name[50];
    float marks;
};
int main() {
    struct Student s;
    printf("Enter Roll Number: ");
    scanf("%d", &s.roll);
    printf("Enter Name: ");
    scanf("%s", s.name);
    printf("Enter Marks: ");
    scanf("%f", &s.marks);
    printf("\nStudent Details:\n");
    printf("Roll: %d\nName: %s\nMarks: %.2f\n", s.roll, s.name, s.marks);
    return 0;
}
```

2. Store Employee Details Using Structure

```
#include <stdio.h>
struct Employee {
    int id;
    char name[50];
    float salary;
};
int main() {
    struct Employee e;
    printf("Enter Employee ID: ");
    scanf("%d", &e.id);
    printf("Enter Employee Name: ");
    scanf("%s", e.name);
    printf("Enter Salary: ");
    scanf("%f", &e.salary);
    printf("\nEmployee Details:\nID: %d\nName: %s\nSalary: %.2f\n", e.id,
e.name, e.salary);
}
```

```
    return 0;
}
```

3. Nested Structures Example

```
#include <stdio.h>
struct Date {
    int day;
    int month;
    int year;
};
struct Student {
    int roll;
    char name[50];
    struct Date dob;
};
int main() {
    struct Student s;
    s.roll = 101;
    printf("Enter Name: ");
    scanf("%s", s.name);
    printf("Enter Date of Birth (dd mm yyyy): ");
    scanf("%d %d %d", &s.dob.day, &s.dob.month, &s.dob.year);
    printf("\nStudent Details:\nRoll: %d\nName: %s\nDOB: %02d-%02d-%04d\n",
s.roll, s.name, s.dob.day, s.dob.month, s.dob.year);
    return 0;
}
```

4. Store Student Data in a File

```
#include <stdio.h>
struct Student {
    int roll;
    char name[50];
    float marks;
};
int main() {
    struct Student s;
    FILE *fp;
    fp = fopen("student.txt", "w");
    if(fp == NULL) { printf("Error opening file!"); return 1; }
    printf("Enter Roll: "); scanf("%d", &s.roll);
    printf("Enter Name: "); scanf("%s", s.name);
    printf("Enter Marks: "); scanf("%f", &s.marks);
    fprintf(fp, "%d %s %.2f\n", s.roll, s.name, s.marks);
}
```

```

    fclose(fp);
    printf("Data stored successfully!\n");
    return 0;
}

```

5. Read Student Records from a File

```

#include <stdio.h>
int main() {
    FILE *fp;
    int roll;
    char name[50];
    float marks;
    fp = fopen("student.txt", "r");
    if(fp == NULL) { printf("File not found!\n"); return 1; }
    printf("Student Records:\n");
    while(fscanf(fp, "%d %s %f", &roll, name, &marks) != EOF) {
        printf("Roll: %d, Name: %s, Marks: %.2f\n", roll, name, marks);
    }
    fclose(fp);
    return 0;
}

```

6. Append Data to a File

```

#include <stdio.h>
int main() {
    FILE *fp;
    int roll;
    char name[50];
    float marks;
    fp = fopen("student.txt", "a");
    if(fp == NULL) { printf("Error opening file!"); return 1; }
    printf("Enter Roll: "); scanf("%d", &roll);
    printf("Enter Name: "); scanf("%s", name);
    printf("Enter Marks: "); scanf("%f", &marks);
    fprintf(fp, "%d %s %.2f\n", roll, name, marks);
    fclose(fp);
    printf("Data appended successfully!\n");
    return 0;
}

```

7. Copy Contents of One File to Another

```
#include <stdio.h>
int main() {
    FILE *src, *dest;
    char ch;
    src = fopen("student.txt", "r");
    dest = fopen("backup.txt", "w");
    if(src == NULL || dest == NULL) { printf("Error opening file!\n"); return
1; }
    while((ch = fgetc(src)) != EOF) { fputc(ch, dest); }
    fclose(src); fclose(dest);
    printf("File copied successfully!\n");
    return 0;
}
```

8. Count Characters in a File

```
#include <stdio.h>
int main() {
    FILE *fp;
    char ch;
    int count = 0;
    fp = fopen("student.txt", "r");
    if(fp == NULL) { printf("File not found!\n"); return 1; }
    while((ch = fgetc(fp)) != EOF) { count++; }
    fclose(fp);
    printf("Total characters: %d\n", count);
    return 0;
}
```

9. Count Words in a File

```
#include <stdio.h>
int main() {
    FILE *fp;
    char ch;
    int words = 0;
    fp = fopen("student.txt", "r");
    if(fp == NULL) { printf("File not found!\n"); return 1; }
    while((ch = fgetc(fp)) != EOF) {
        if(ch == ' ' || ch == '\n' || ch == '\t') words++;
    }
    fclose(fp);
}
```

```

    printf("Total words: %d\n", words);
    return 0;
}

```

10. Count Lines in a File

```

#include <stdio.h>
int main() {
    FILE *fp;
    char ch;
    int lines = 0;
    fp = fopen("student.txt", "r");
    if(fp == NULL) { printf("File not found!\n"); return 1; }
    while((ch = fgetc(fp)) != EOF) {
        if(ch == '\n') lines++;
    }
    fclose(fp);
    printf("Total lines: %d\n", lines);
    return 0;
}

```

11. Store 100 Employee Records in a File

```

#include <stdio.h>
struct Employee { int id; char name[50]; float salary; };
int main() {
    struct Employee e;
    FILE *fp = fopen("employee.txt", "w");
    if(fp == NULL) return 1;
    for(int i = 0; i < 100; i++) {
        printf("Enter ID, Name, Salary: ");
        scanf("%d %s %f", &e.id, e.name, &e.salary);
        fprintf(fp, "%d %s %.2f\n", e.id, e.name, e.salary);
    }
    fclose(fp);
    printf("100 employee records stored.\n");
    return 0;
}

```

12. Search a Record in a File

```

#include <stdio.h>
struct Employee { int id; char name[50]; float salary; };
int main() {

```

```

FILE *fp = fopen("employee.txt", "r");
struct Employee e;
int id, found = 0;
if(fp == NULL) return 1;
printf("Enter ID to search: "); scanf("%d", &id);
while(fscanf(fp, "%d %s %f", &e.id, e.name, &e.salary) != EOF) {
    if(e.id == id) { printf("Record Found: %d %s %.2f\n", e.id, e.name,
e.salary); found = 1; break; }
}
if(!found) printf("Record not found.\n");
fclose(fp);
return 0;
}

```

13. Update a Record in a File

```

#include <stdio.h>
struct Employee { int id; char name[50]; float salary; };
int main() {
    FILE *fp = fopen("employee.txt", "r"), *temp = fopen("temp.txt", "w");
    struct Employee e;
    int id, found = 0;
    if(fp == NULL || temp == NULL) return 1;
    printf("Enter ID to update: "); scanf("%d", &id);
    while(fscanf(fp, "%d %s %f", &e.id, e.name, &e.salary) != EOF) {
        if(e.id == id) { printf("Enter new Name and Salary: "); scanf("%s %f",
e.name, &e.salary); found = 1; }
        fprintf(temp, "%d %s %.2f\n", e.id, e.name, e.salary);
    }
    fclose(fp); fclose(temp);
    remove("employee.txt"); rename("temp.txt", "employee.txt");
    if(found) printf("Record updated.\n"); else printf("Record not found.\n");
    return 0;
}

```

14. Delete a Record from a File

```

#include <stdio.h>
struct Employee { int id; char name[50]; float salary; };
int main() {
    FILE *fp = fopen("employee.txt", "r"), *temp = fopen("temp.txt", "w");
    struct Employee e;
    int id, found = 0;
    if(fp == NULL || temp == NULL) return 1;
    printf("Enter ID to delete: "); scanf("%d", &id);

```

```
while(fscanf(fp, "%d %s %f", &e.id, e.name, &e.salary) != EOF) {  
    if(e.id == id) { found = 1; continue; }  
    fprintf(temp, "%d %s %.2f\n", e.id, e.name, e.salary);  
}  
fclose(fp); fclose(temp);  
remove("employee.txt"); rename("temp.txt", "employee.txt");  
if(found) printf("Record deleted.\n"); else printf("Record not found.\n");  
return 0;  
}
```